TASK 3

ANFAL TARIQ

Cybersecurity

Detailed Report on Ransomware Attacks

1. RansomwareAttack
I have chosen the **Ransomware attack**. This is one of the most dangerous and common cyber threats today. It has affected hospitals, schools, businesses, and even government agencies around the world.

2. Explaination of the Attack
A ransomware attack is a type of malicious software (malware) that blocks access to a victim's files or computer system. The attacker then demands a payment, or "ransom," in exchange for giving back access.

How the attack works:
First, the attacker must get the ransomware into the victim's system. They usually do this through **phishing emails**. For example, an employee might receive an email that looks like an important invoice. The email contains a link or an attachment. When the employee clicks the link or opens the attachment, the ransomware program is silently downloaded and installed. Another common way attackers gain access is through **Remote Desktop Protocol (RDP)** . If a company has a weak password for its remote access, hackers can guess it and install the ransomware directly.

Once inside the system, the ransomware works very quickly. It scans the computer and network for files. It then uses a strong encryption algorithm to lock these files. Encryption turns normal files into scrambled, unreadable code. The victim will see a message on their screen telling them that their files are locked. The message gives instructions on how to pay the ransom, usually in cryptocurrency like Bitcoin because it is difficult to trace. If the victim does not pay by a certain deadline, the hackers threaten to delete the files forever.

Ransomware usually affects critical business documents, databases, customer records, and backup files. In severe cases, it can shut down entire hospital systems, preventing doctors from accessing patient records.

3. Impact Analysis
The damage caused by a ransomware attack can be devastating.

- **Data Loss:** Even if a victim pays the ransom, there is no guarantee the hackers will unlock the files. Sometimes the decryption tool fails, resulting in permanent loss of important photos, financial records, and legal documents.
- **Financial Loss:** The damage is not just about the ransom fee. Companies often lose millions of dollars because their business stops completely during the attack. This is called "downtime." They also have to pay expensive IT experts to clean up the systems and restore data.
- **Privacy Issues:** Many modern ransomware attacks also steal data before encrypting it. This is called "double extortion." The attackers threaten to publish sensitive customer information, like names, addresses, and credit card numbers, if the ransom is not paid. This violates privacy laws and puts customers at risk of identity theft.

- **Damage to Trust and Reputation:** When a company suffers a ransomware attack, customers lose trust. If a bank or a hospital cannot protect its data, people will take their business elsewhere. Rebuilding a good reputation after a severe cyberattack can take many years.

4. Security Measures

Preventing a ransomware attack is much better than dealing with the aftermath.

- **Regular Data Backups:** This is the most effective defense. Companies should follow the "3-2-1 rule." Keep three copies of your data, stored on two different types of media, and one copy stored off-site or in the cloud. If ransomware hits, you can simply wipe your system and restore your clean backup without paying the ransom.
- **Keep Software Updated:** Attackers often exploit known weaknesses in old software. Make sure your operating system, web browsers, and antivirus programs are always updated to the latest versions. These updates include security patches that fix known holes.
- **Use Strong Endpoint Protection:** Install a reliable antivirus and anti-malware program on all devices. These tools can detect suspicious behavior and block ransomware before it can run.
- **Employee Training and Awareness:** Since phishing emails are the number one delivery method, training employees is critical. Teach staff to recognize suspicious emails, avoid opening unknown attachments, and never click on links from untrusted sources.
- **Restrict User Permissions:** Not every employee needs full access to every file. Implement the "principle of least privilege." This means giving users only the access they need to do their jobs. If a user's account is compromised, the attacker will have limited access to the network.
- **Disable Macros in Office Documents:** Many ransomware strains are delivered through malicious macros in Word or Excel files. Disabling macros by default can stop the infection process.

5. Conclusion

It is a powerful and dangerous attack that can cripple an organization in minutes. It causes serious harm through data loss, financial ruin, privacy violations, and the destruction of hard-earned trust. However, these attacks are not unstoppable.By staying informed, practicing caution, and investing in basic security tools like backups and updates, we can make ourselves a much harder target.